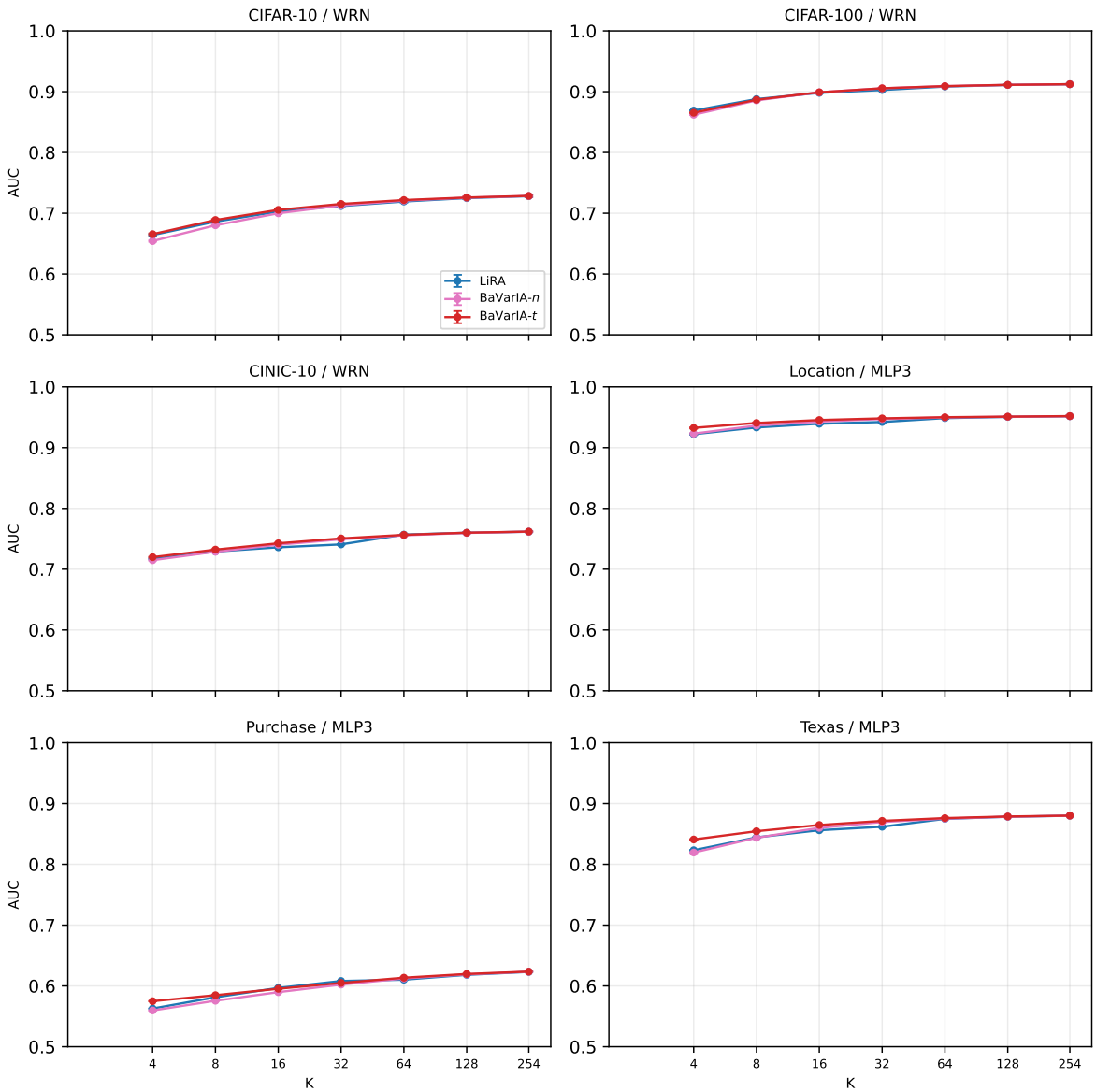

Figure 1: Online AUC vs. shadow-model budget $K \in \{4, 8, 16, 32, 64, 128, 254\}$, WRN / MLP3 testbeds. LiRA, BaVarIA-n (NIG-shrinkage variance, normal posterior predictive), BaVarIA-t (NIG-shrinkage variance, t posterior predictive). BaVarIA replaces LiRA's pooled-vs-per-sample variance switch with a single shrinkage estimator continuous in K. Mean ± 1 SE over 32 leave-one-out replicates, antithetic-paired design (128 pairs, $K = 254$ max). Online setting uses both IN and OUT shadow observations.

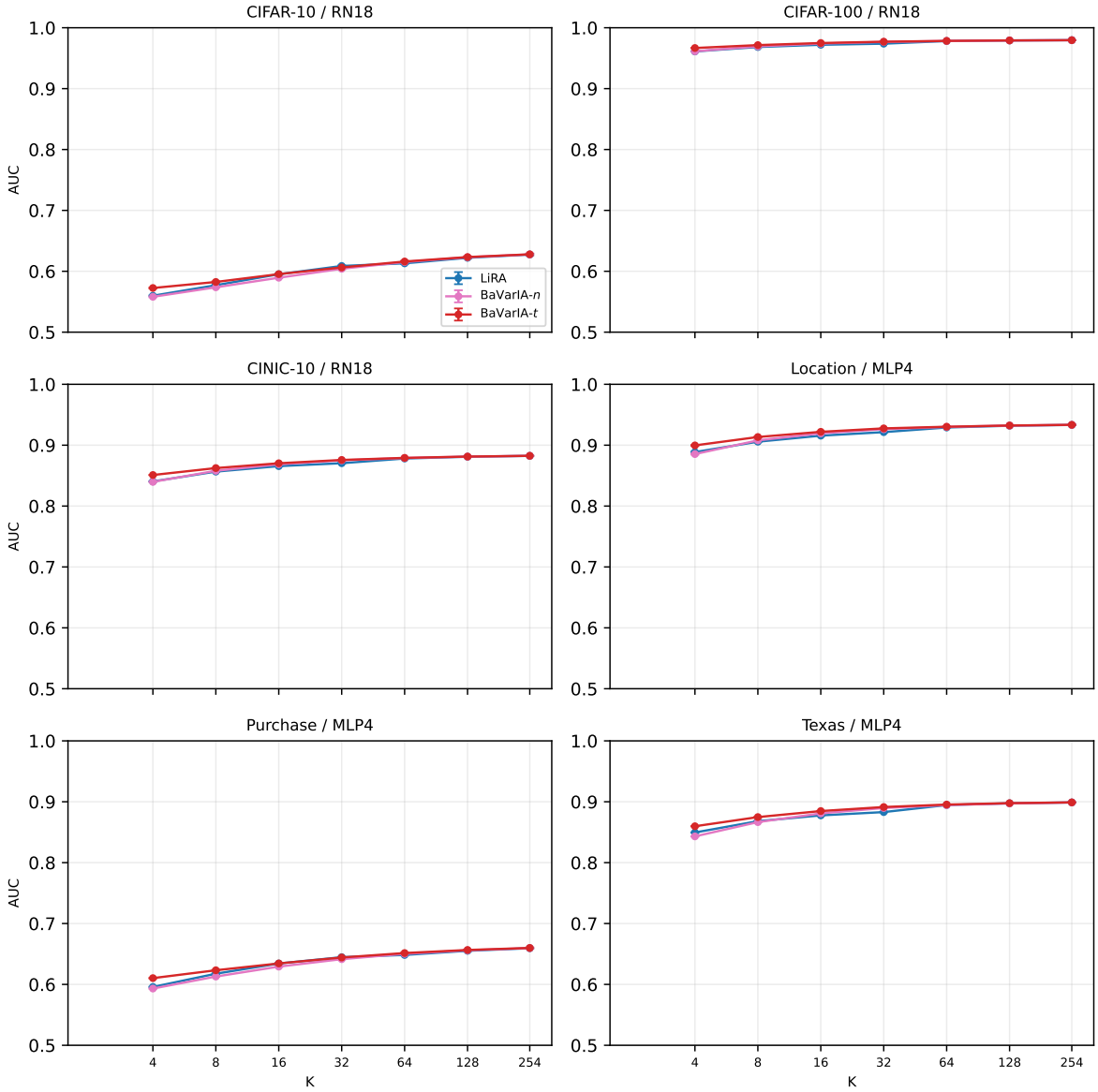

Figure 2: Online AUC vs. K, RN18 / MLP4 testbeds. LiRA, BaVarIA-n, BaVarIA-t. Mean ± 1 SE over 32 replicates.

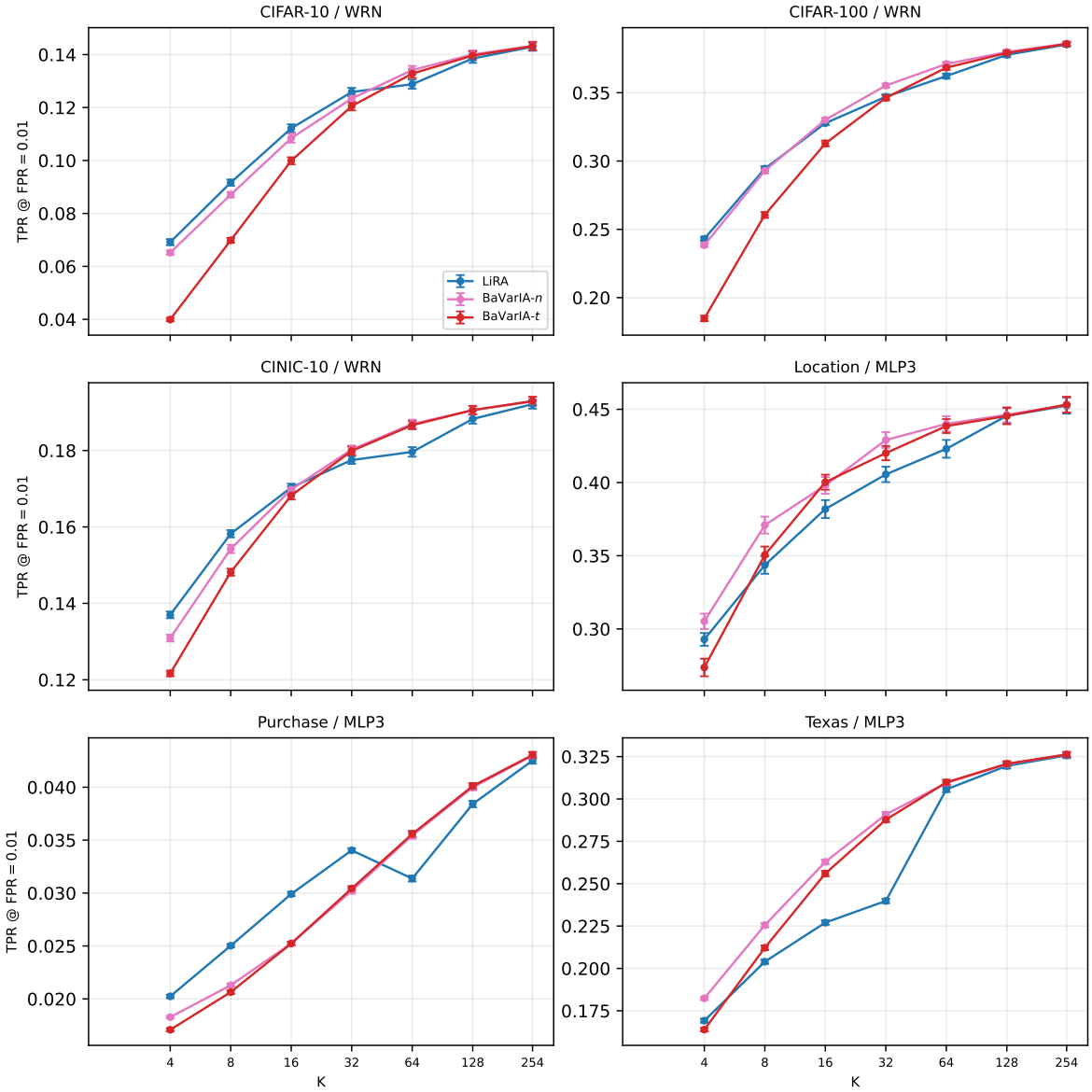

Figure 3: Online TPR at FPR= 0.01 vs. K, WRN / MLP3 testbeds. LiRA, BaVarIA-n, BaVarIA-t. Mean ± 1 SE over 32 replicates.

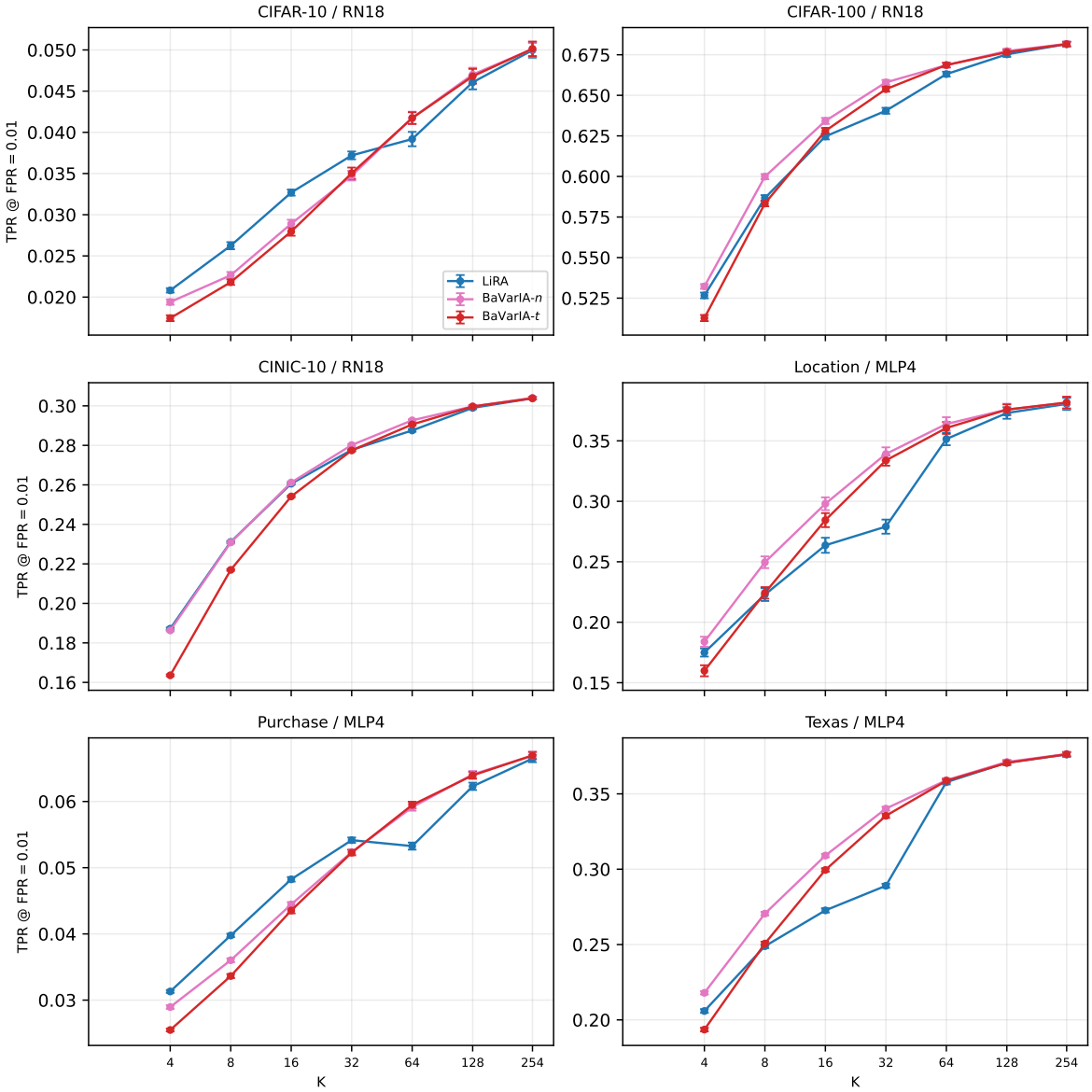

Figure 4: Online TPR at FPR= 0.01 vs. K, RN18 / MLP4 testbeds. LiRA, BaVarIA-n, BaVarIA-t. Mean ± 1 SE over 32 replicates.

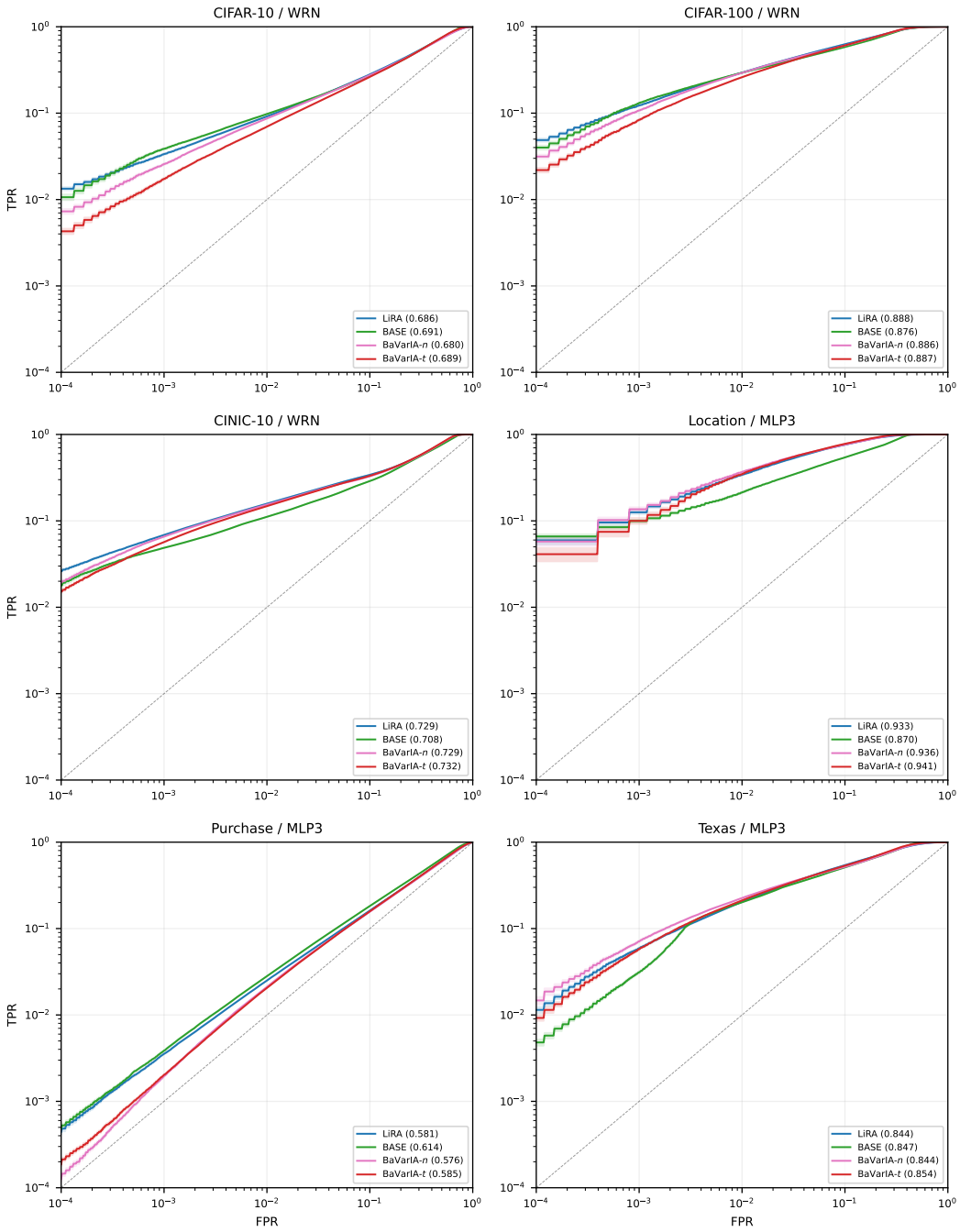

Figure 5: Online ROC at $K = 8$, WRN / MLP3 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid; ROC curves interpolated onto a common log-spaced FPR grid and aggregated as the mean over 32 replicates, ± 1 SE bands.

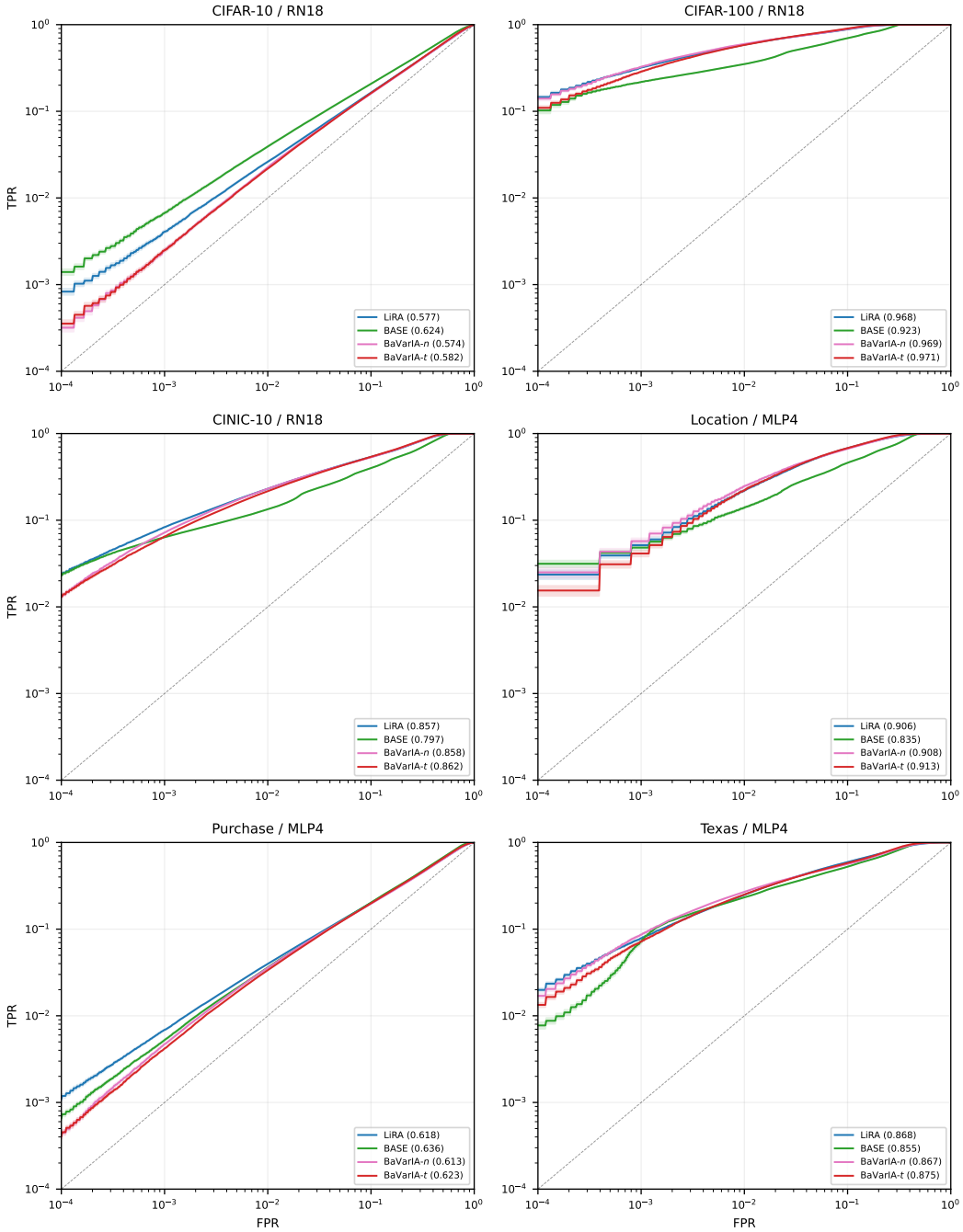

Figure 6: Online ROC at $K = 8$, RN18 / MLP4 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

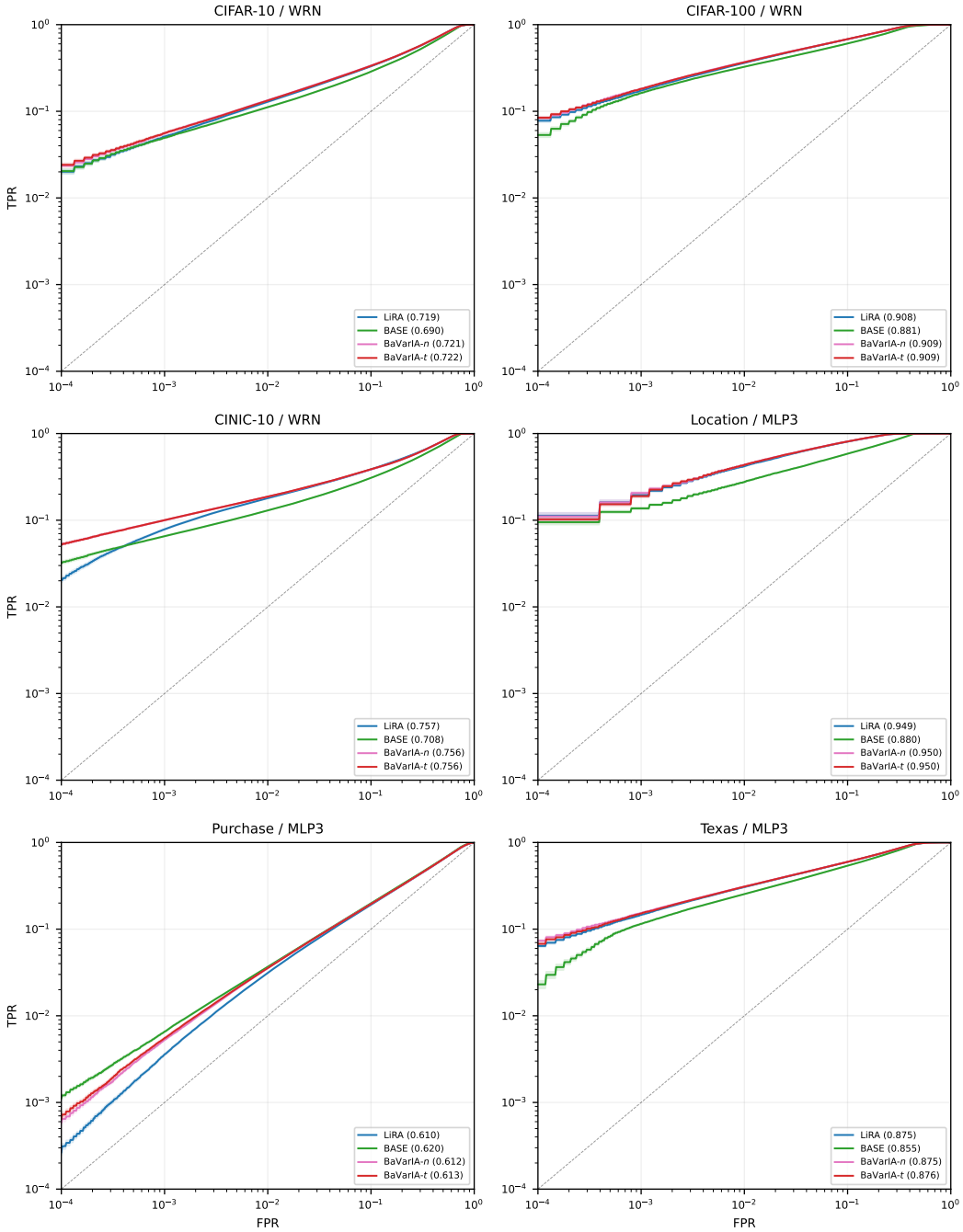

Figure 7: Online ROC at $K = 64$, WRN / MLP3 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

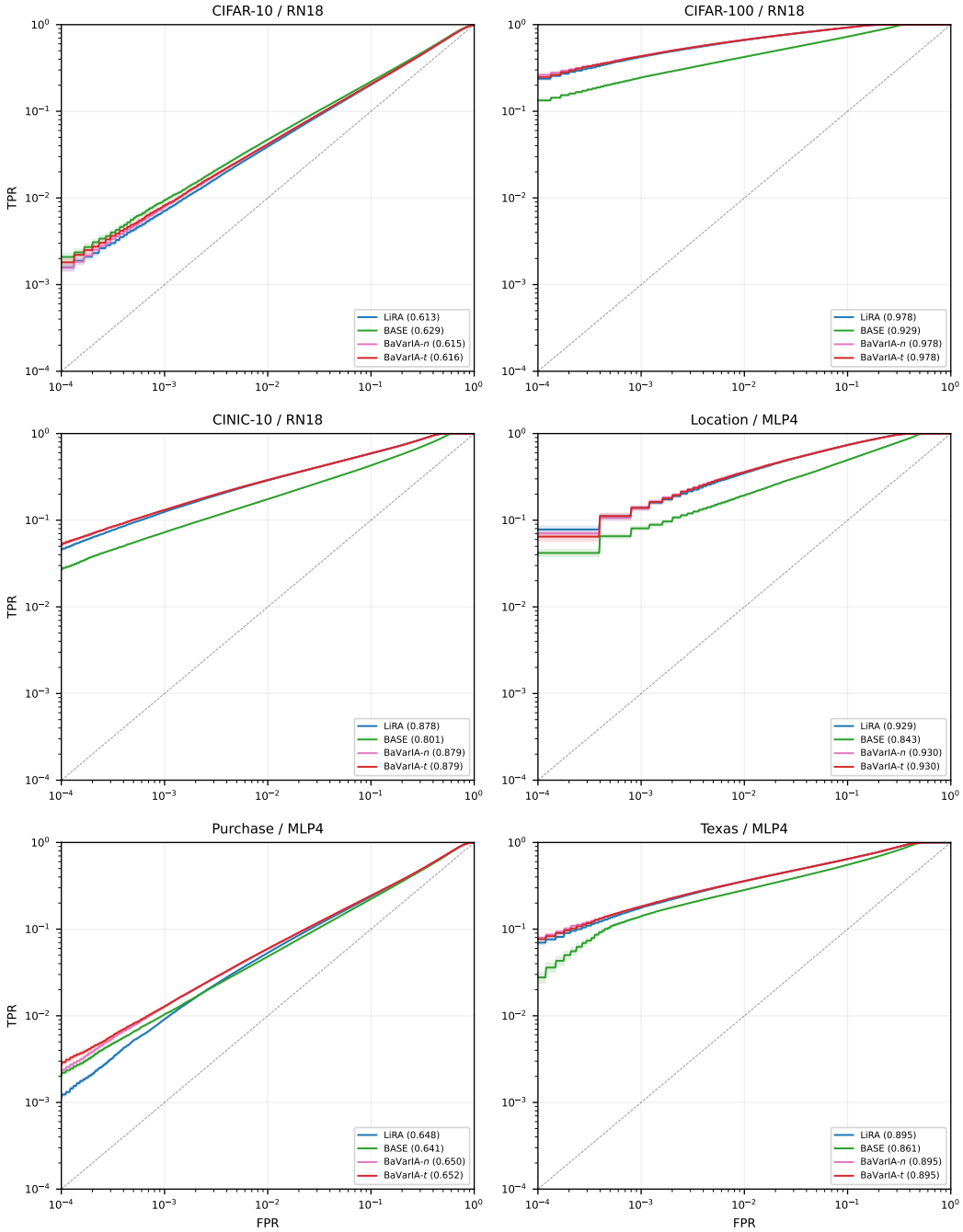

Figure 8: Online ROC at $K = 64$, RN18 / MLP4 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

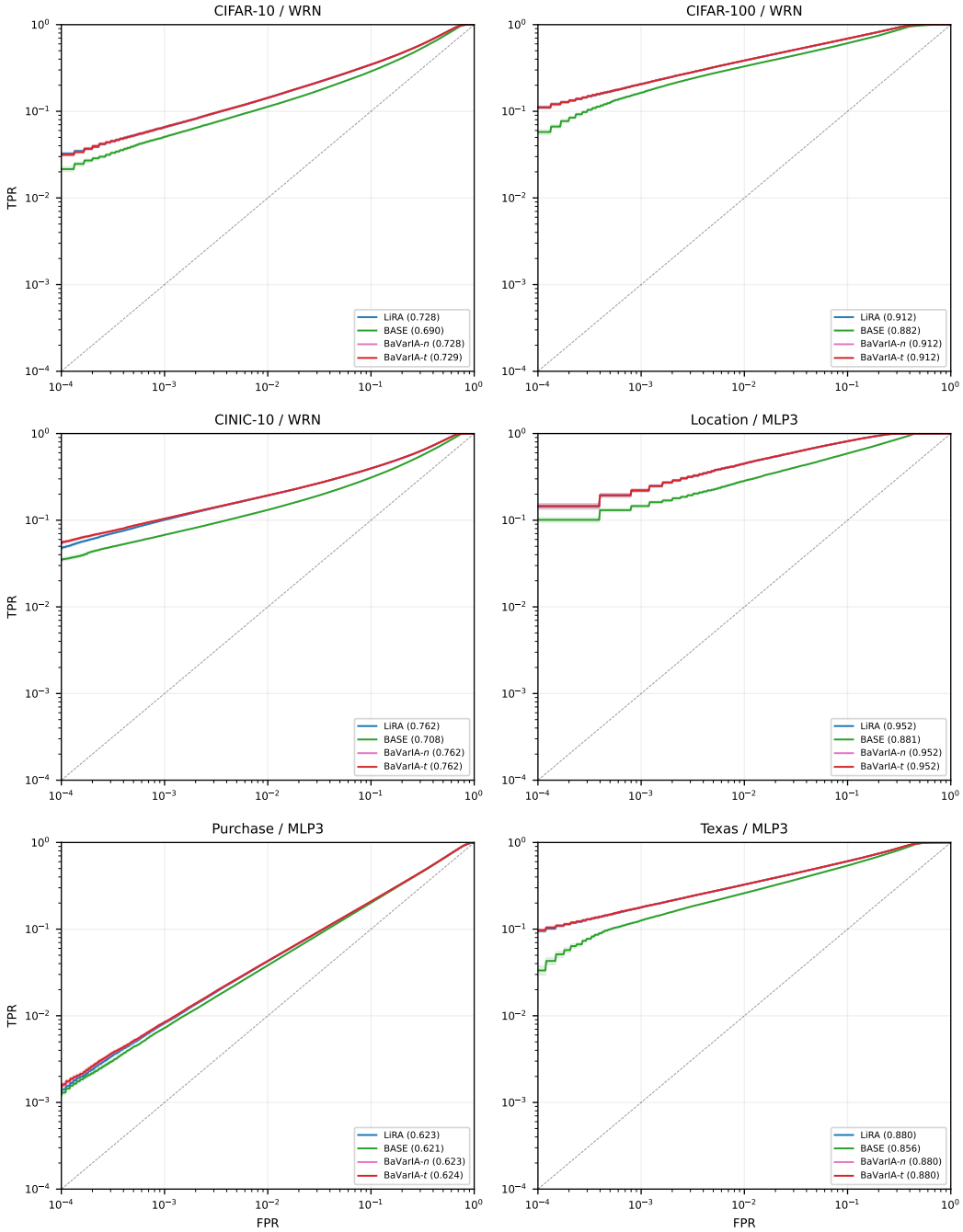

Figure 9: Online ROC at $K = 254$, WRN / MLP3 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

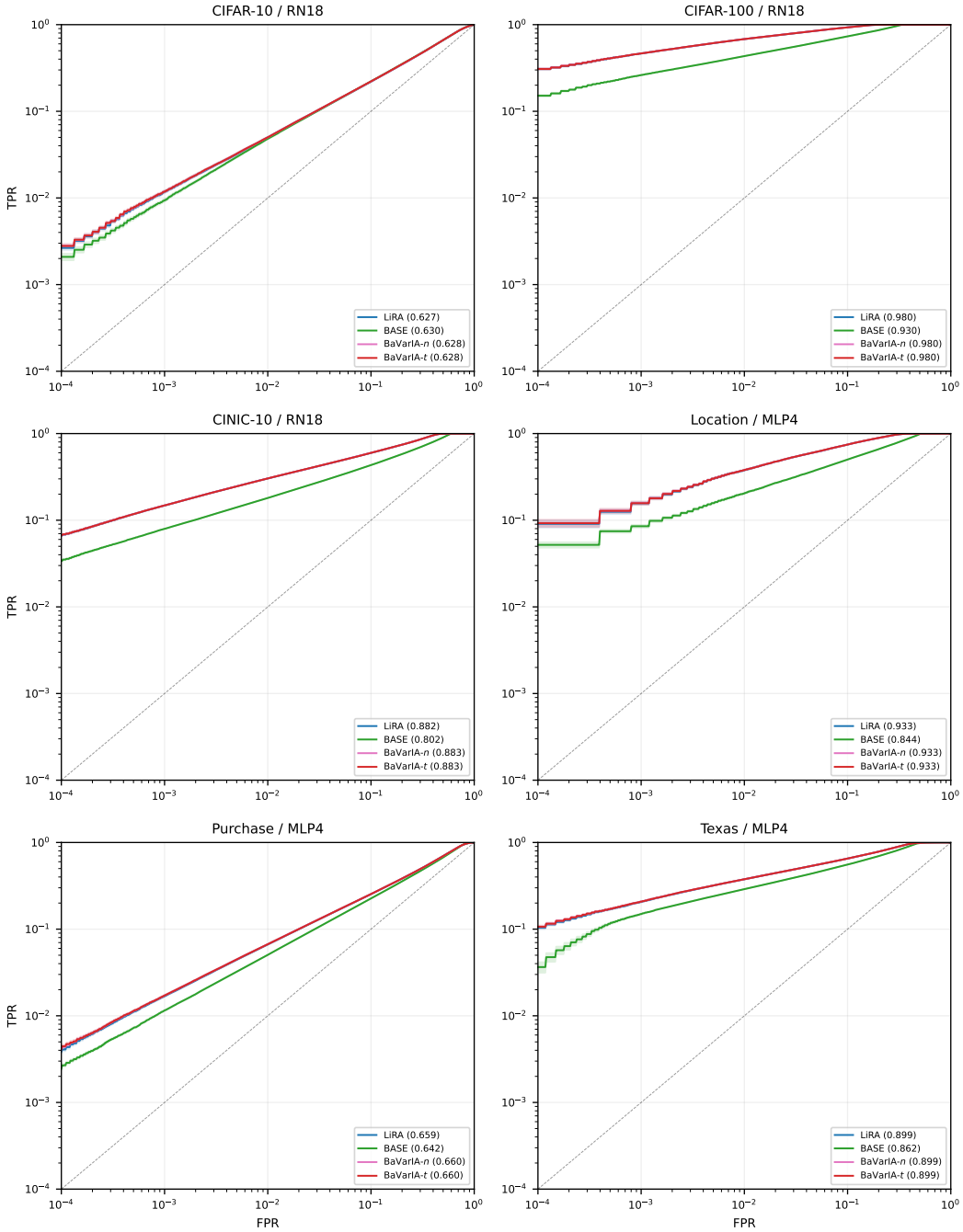

Figure 10: Online ROC at $K = 254$, RN18 / MLP4 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	0.686	0.680	**0.689**
CIFAR-100 / WRN	**0.888**	0.886	0.887
CINIC-10 / WRN	0.729	0.729	**0.732**
Location / MLP3	0.933	0.936	**0.941**
Purchase / MLP3	0.581	0.576	**0.585**
Texas / MLP3	0.844	0.844	**0.854**
CIFAR-10 / RN18	0.577	0.574	**0.582**
CIFAR-100 / RN18	0.968	0.969	**0.971**
CINIC-10 / RN18	0.857	0.858	**0.862**
Location / MLP4	0.906	0.908	**0.913**
Purchase / MLP4	0.618	0.613	**0.623**
Texas / MLP4	0.868	0.867	**0.875**

Table 1: Online AUC, $K = 8$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	0.719	0.721	**0.722**
CIFAR-100 / WRN	0.908	**0.909**	**0.909**
CINIC-10 / WRN	**0.757**	0.756	0.756
Location / MLP3	0.949	0.950	**0.950**
Purchase / MLP3	0.610	0.612	**0.613**
Texas / MLP3	0.875	0.875	**0.876**
CIFAR-10 / RN18	0.613	0.615	**0.616**
CIFAR-100 / RN18	0.978	0.978	**0.978**
CINIC-10 / RN18	0.878	0.879	**0.879**
Location / MLP4	0.929	0.930	**0.930**
Purchase / MLP4	0.648	0.650	**0.652**
Texas / MLP4	0.895	0.895	**0.895**

Table 2: Online AUC, $K = 64$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	0.728	0.728	**0.729**
CIFAR-100 / WRN	**0.912**	**0.912**	**0.912**
CINIC-10 / WRN	0.762	0.762	**0.762**
Location / MLP3	0.952	0.952	**0.952**
Purchase / MLP3	0.623	0.623	**0.624**
Texas / MLP3	0.880	0.880	**0.880**
CIFAR-10 / RN18	0.627	0.628	**0.628**
CIFAR-100 / RN18	**0.980**	**0.980**	**0.980**
CINIC-10 / RN18	**0.882**	**0.883**	**0.883**
Location / MLP4	0.933	0.933	**0.933**
Purchase / MLP4	0.659	0.660	**0.660**
Texas / MLP4	**0.899**	0.899	**0.899**

Table 3: Online AUC, $K = 254$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.092**	0.087	0.070
CIFAR-100 / WRN	**0.294**	0.293	0.261
CINIC-10 / WRN	**0.158**	0.154	0.148
Location / MLP3	0.344	**0.371**	0.351
Purchase / MLP3	**0.025**	0.021	0.021
Texas / MLP3	0.204	**0.226**	0.212
CIFAR-10 / RN18	**0.026**	0.023	0.022
CIFAR-100 / RN18	0.587	**0.600**	0.583
CINIC-10 / RN18	**0.231**	0.231	0.217
Location / MLP4	0.223	**0.250**	0.224
Purchase / MLP4	**0.040**	0.036	0.034
Texas / MLP4	0.249	**0.271**	0.251

Table 4: Online TPR @ FPR= 0.01, $K = 8$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	0.129	**0.134**	0.133
CIFAR-100 / WRN	0.362	**0.371**	0.368
CINIC-10 / WRN	0.180	**0.187**	0.187
Location / MLP3	0.423	**0.440**	0.439
Purchase / MLP3	0.031	0.035	**0.036**
Texas / MLP3	0.306	0.309	**0.310**
CIFAR-10 / RN18	0.039	**0.042**	**0.042**
CIFAR-100 / RN18	0.663	0.669	**0.669**
CINIC-10 / RN18	0.288	**0.293**	0.291
Location / MLP4	0.352	**0.364**	0.361
Purchase / MLP4	0.053	0.059	**0.059**
Texas / MLP4	0.358	**0.359**	0.359

Table 5: Online TPR @ FPR= 0.01, $K = 64$ (LiRA's pooled-vs-per-sample variance-rule cutover). Per-testbed mean over 32 leave-one-out replicates, antithetic-paired design; best per row in bold. BaVarIA-n strictly beats LiRA on 12/12 testbeds, isolating the contribution of NIG-shrinkage variance interpolation. BaVarIA-t ties or trails BaVarIA-n at this K. Average of column means: LiRA 0.266, BaVarIA-n 0.272, BaVarIA-t 0.271.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	0.143	**0.143**	0.143
CIFAR-100 / WRN	0.385	**0.386**	0.386
CINIC-10 / WRN	0.192	**0.193**	**0.193**
Location / MLP3	0.452	0.453	**0.453**
Purchase / MLP3	0.043	**0.043**	**0.043**
Texas / MLP3	0.326	**0.326**	**0.326**
CIFAR-10 / RN18	0.050	**0.050**	**0.050**
CIFAR-100 / RN18	0.682	**0.682**	0.682
CINIC-10 / RN18	0.304	**0.304**	0.304
Location / MLP4	0.381	0.381	**0.382**
Purchase / MLP4	0.066	**0.067**	**0.067**
Texas / MLP4	0.376	**0.376**	0.376

Table 6: Online TPR @ FPR= 0.01, $K = 254$. Mean over 32 replicates. Best per row in bold.

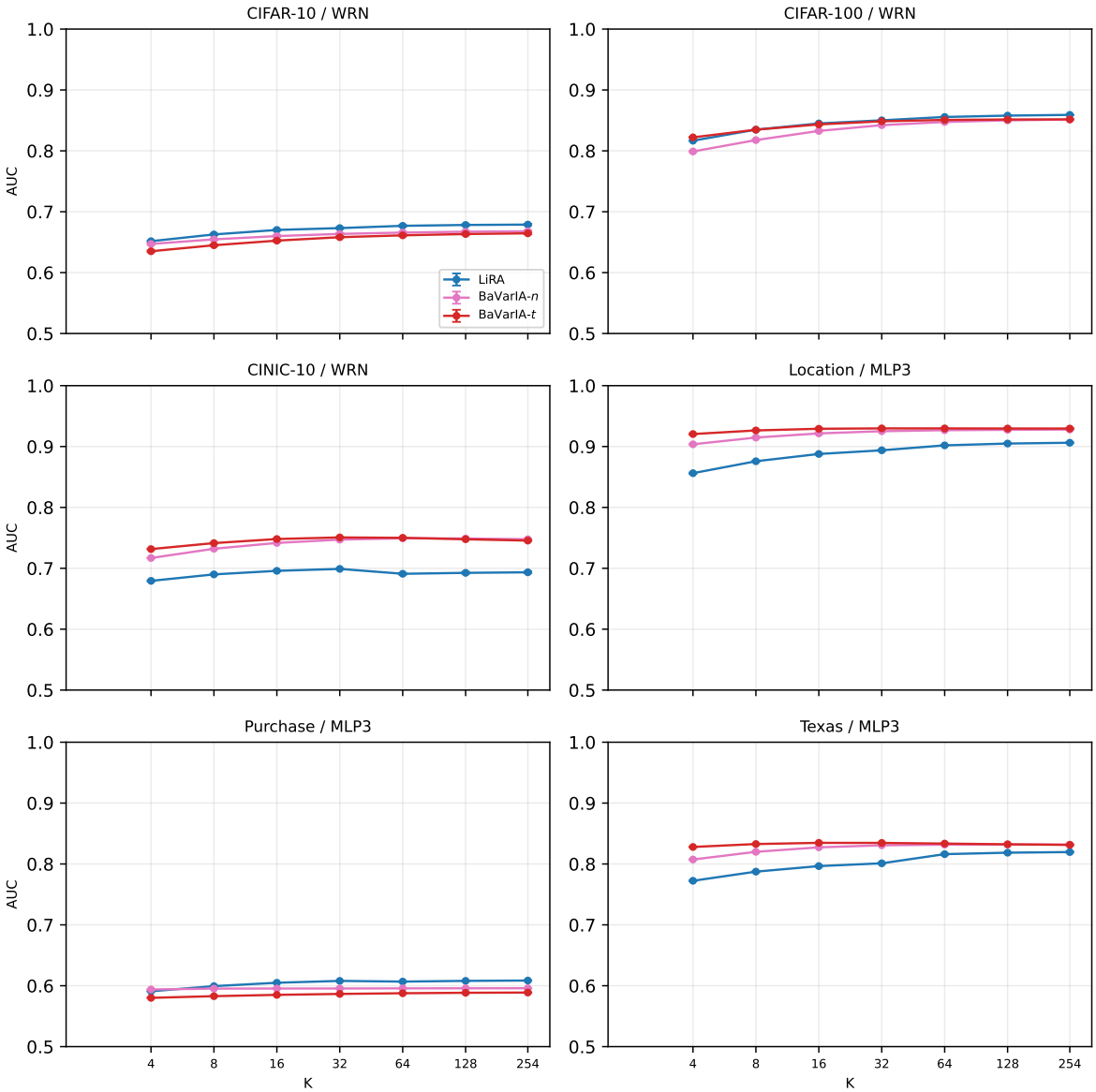

Figure 11: Offline AUC vs. K, WRN / MLP3 testbeds. Offline replaces the IN-class likelihood: LiRA discards it (heuristic rescaling); BaVarIA-n / BaVarIA-t substitute the NIG conjugate prior. Mean ± 1 SE over 32 replicates.

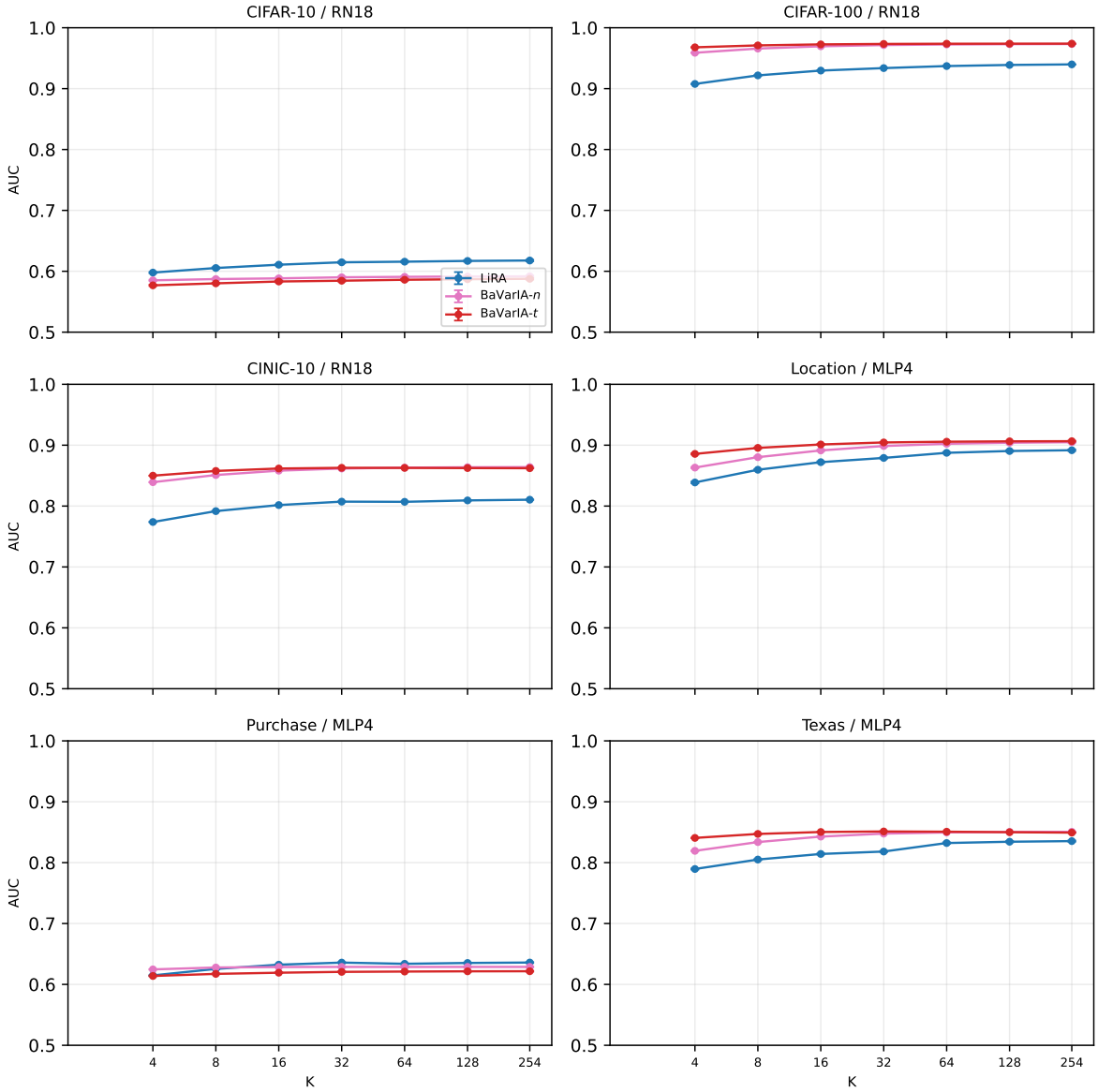

Figure 12: Offline AUC vs. K, RN18 / MLP4 testbeds. LiRA, BaVarIA-n, BaVarIA-t. Mean ± 1 SE over 32 replicates.

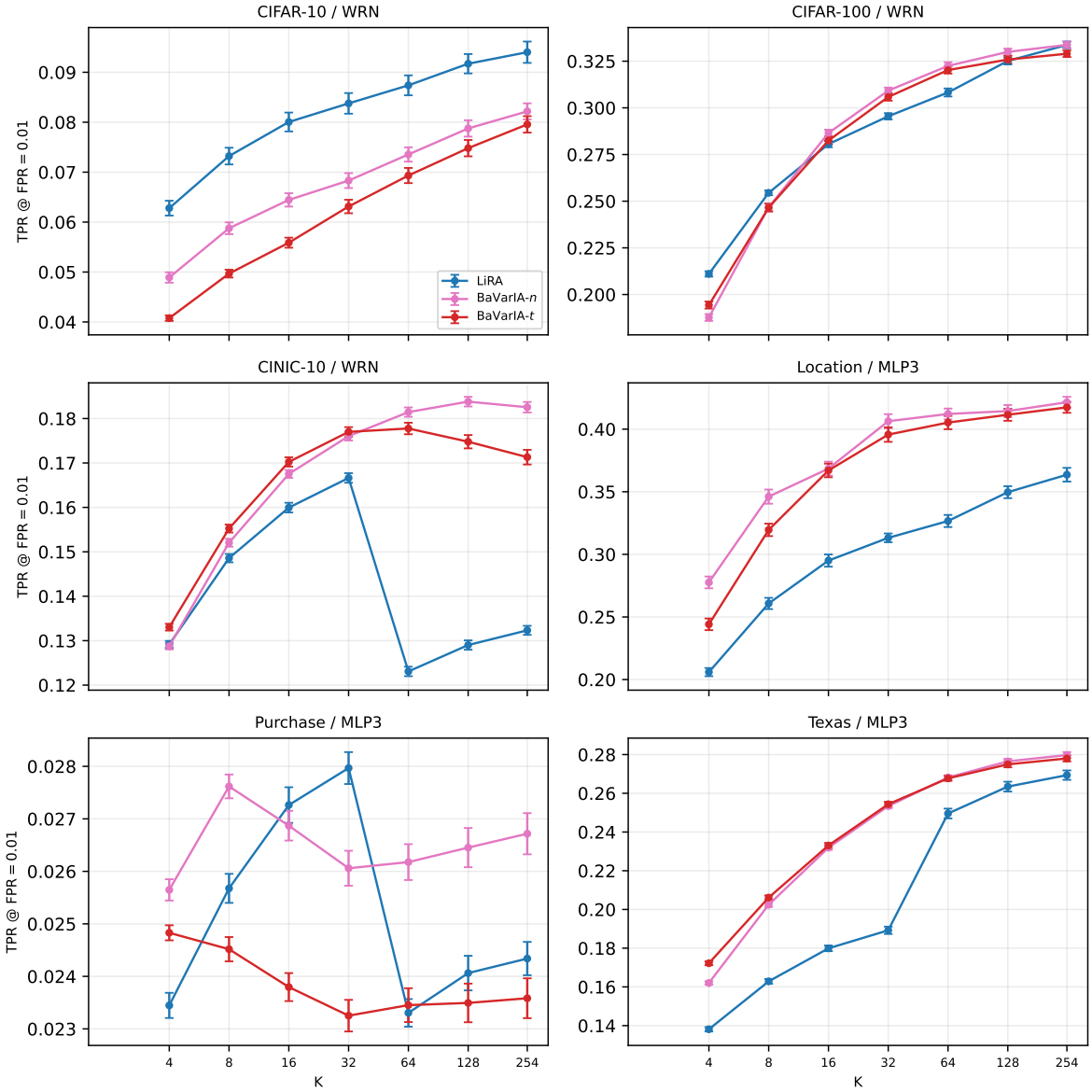

Figure 13: Offline TPR at FPR= 0.01 vs. K, WRN / MLP3 testbeds. LiRA, BaVarIA-n, BaVarIA-t. Mean ± 1 SE over 32 replicates.

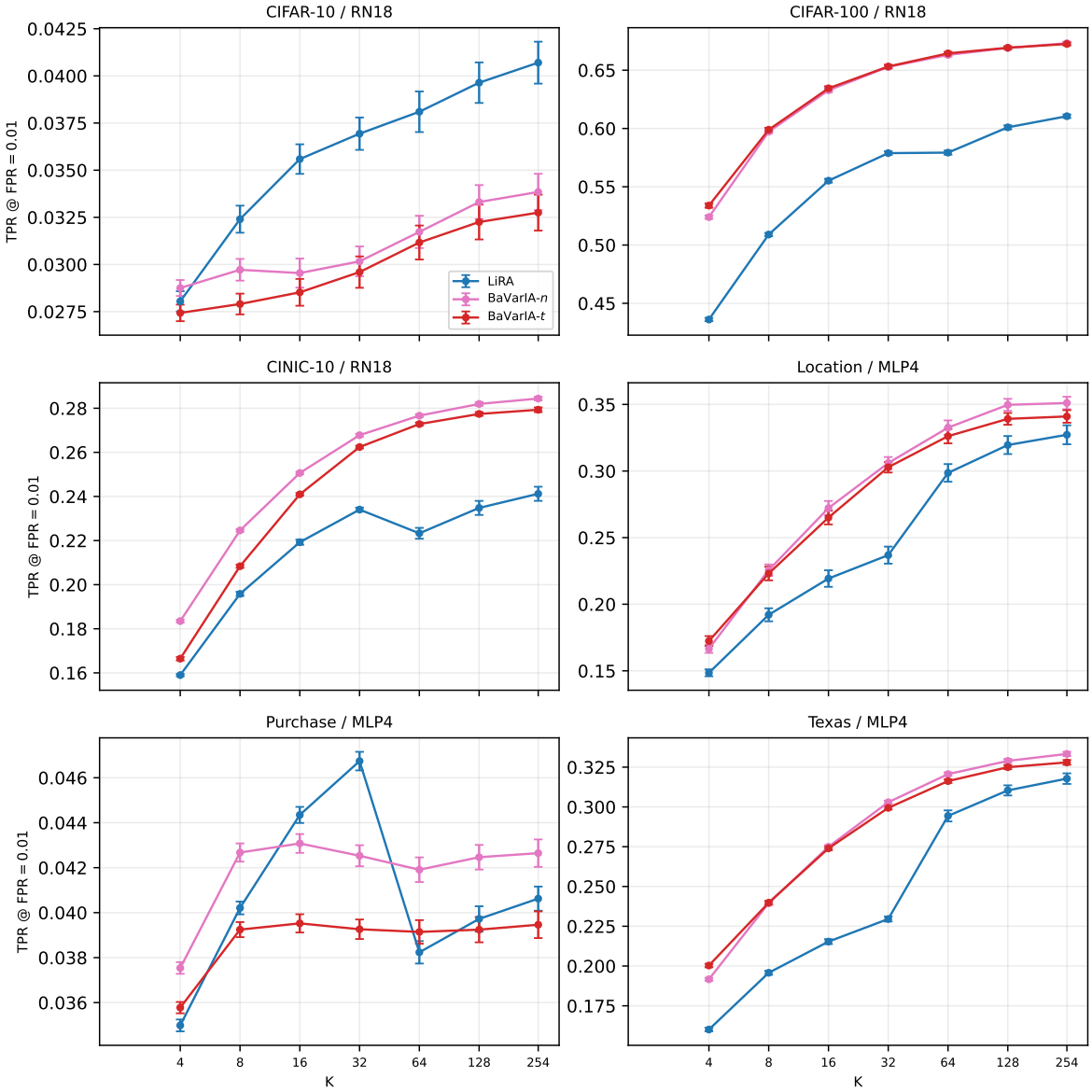

Figure 14: Offline TPR at FPR= 0.01 vs. K, RN18 / MLP4 testbeds. LiRA, BaVarIA-n, BaVarIA-t. Mean ± 1 SE over 32 replicates.

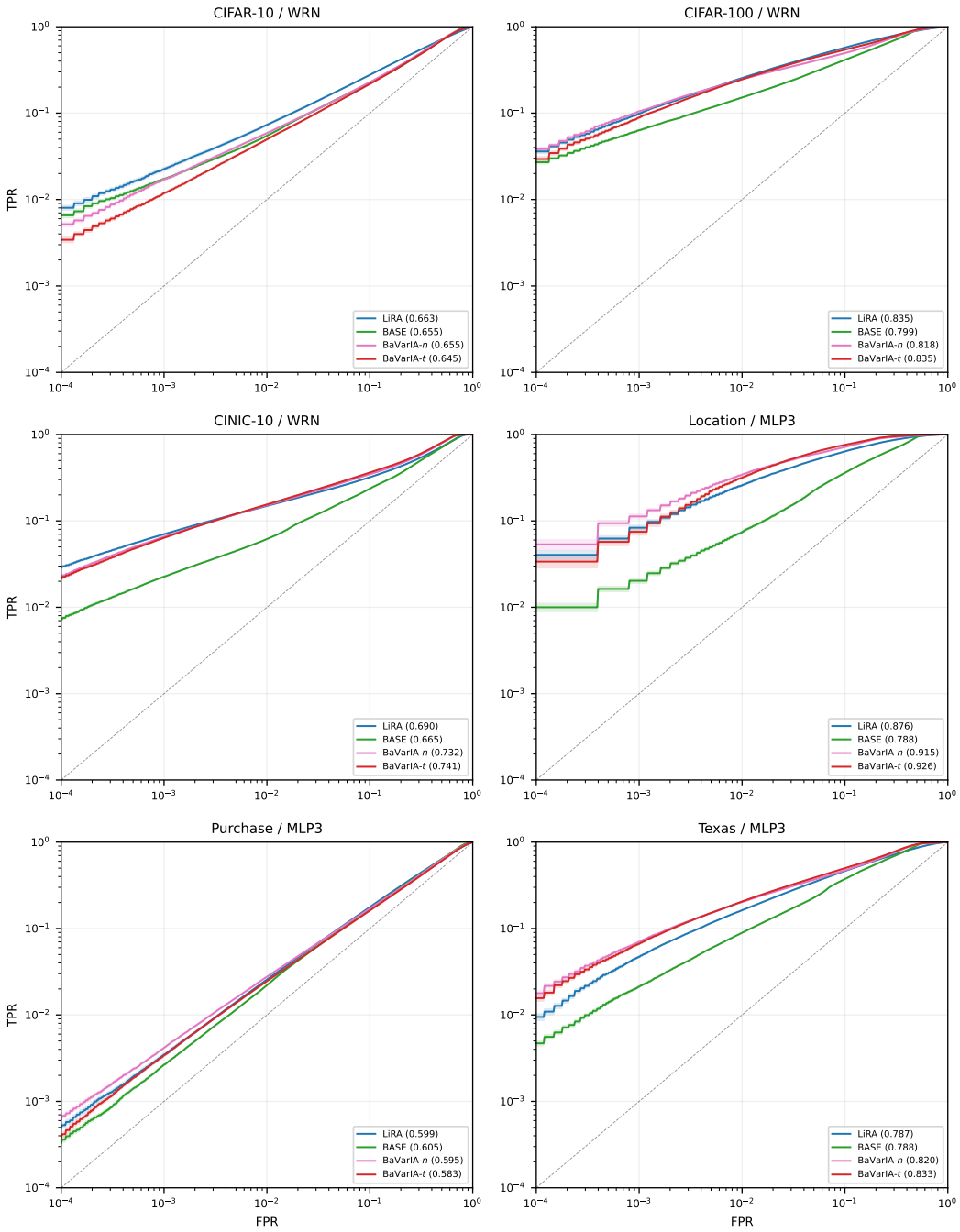

Figure 15: Offline ROC at $K = 8$, WRN / MLP3 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

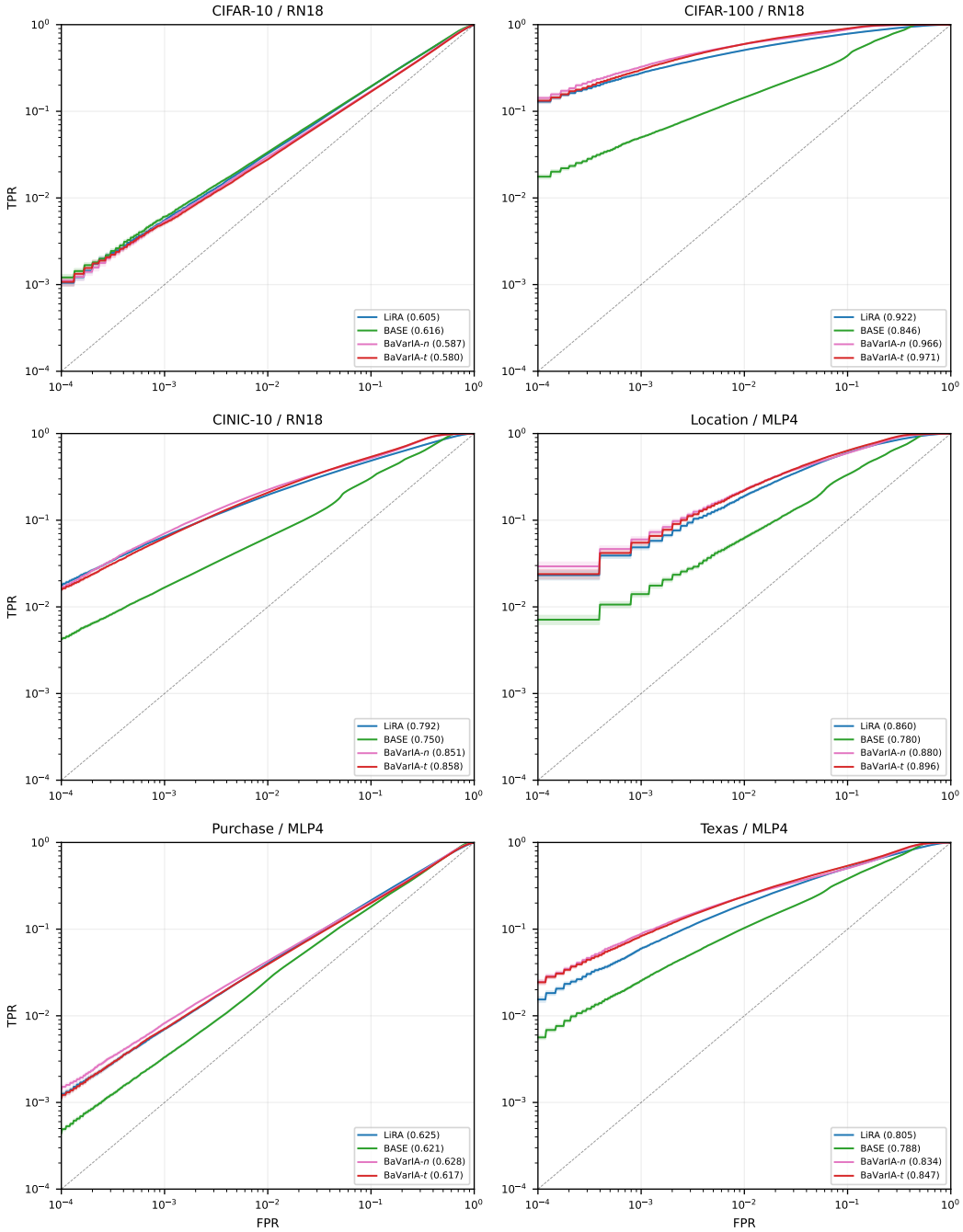

Figure 16: Offline ROC at $K = 8$, RN18 / MLP4 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

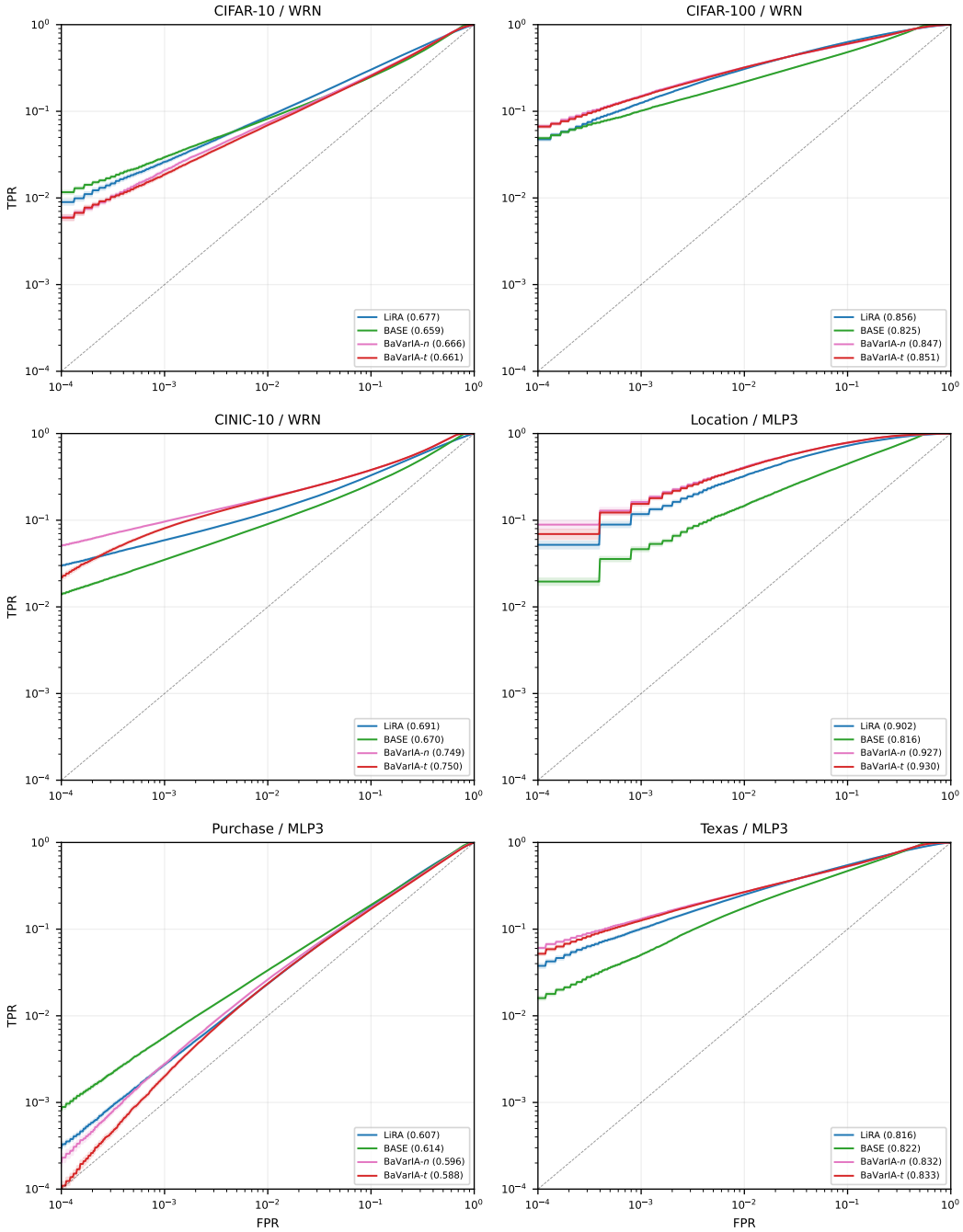

Figure 17: Offline ROC at $K = 64$, WRN / MLP3 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

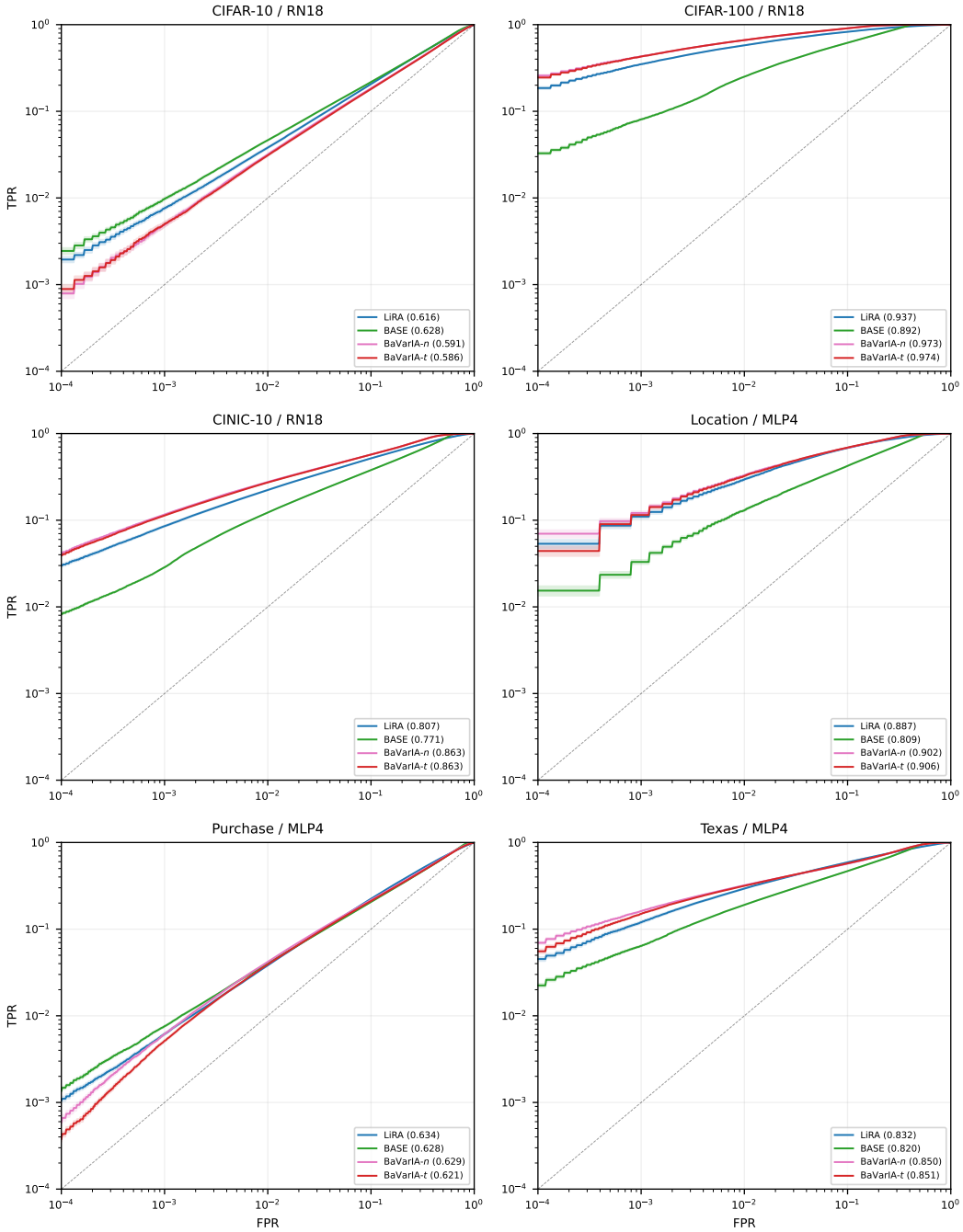

Figure 18: Offline ROC at $K = 64$, RN18 / MLP4 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

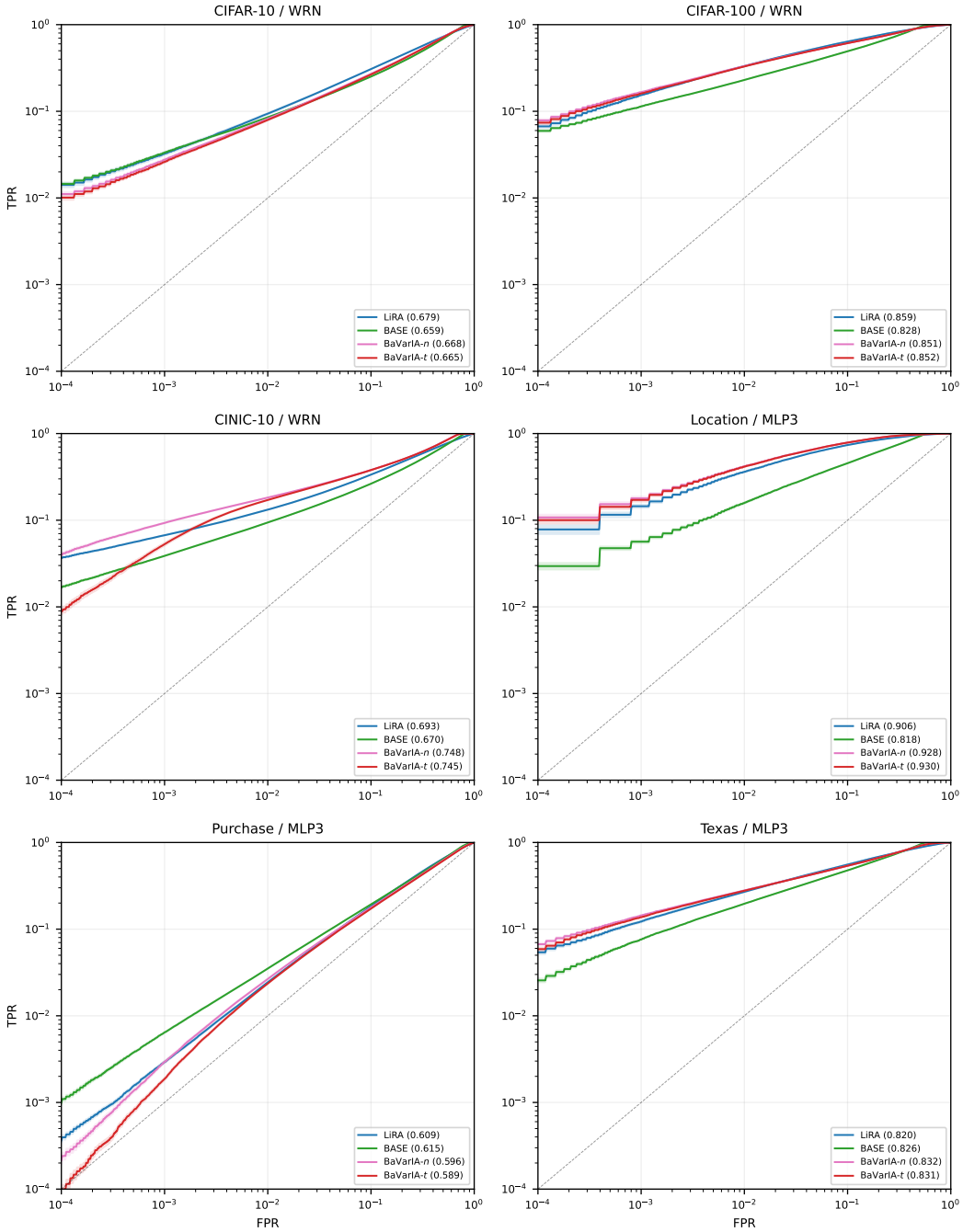

Figure 19: Offline ROC at $K = 254$, WRN / MLP3 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

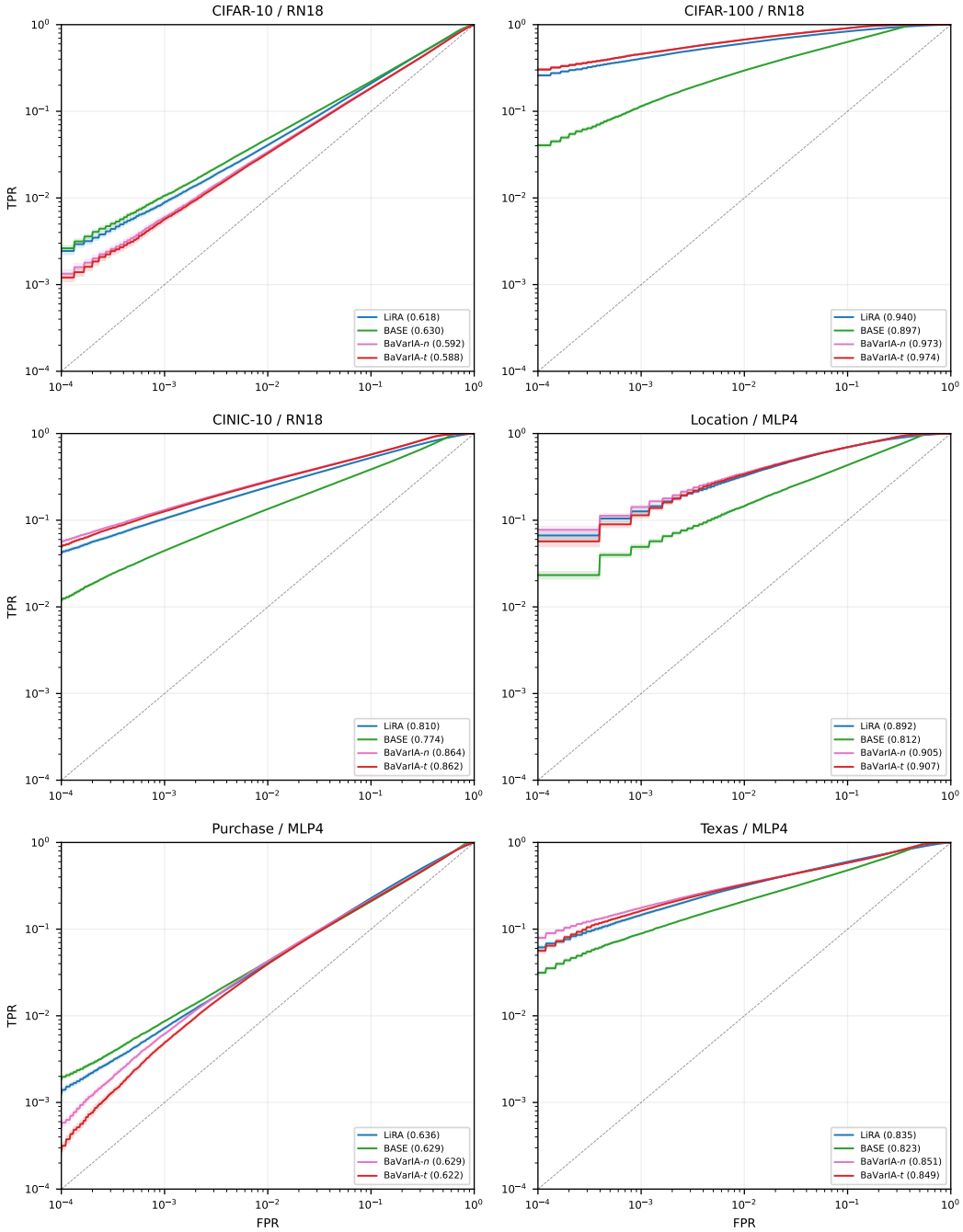

Figure 20: Offline ROC at $K = 254$, RN18 / MLP4 testbeds. LiRA, BASE, BaVarIA-n, BaVarIA-t overlaid.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.663**	0.655	0.645
CIFAR-100 / WRN	**0.835**	0.818	**0.835**
CINIC-10 / WRN	0.690	0.732	**0.741**
Location / MLP3	0.876	0.915	**0.926**
Purchase / MLP3	**0.599**	0.595	0.583
Texas / MLP3	0.787	0.820	**0.833**
CIFAR-10 / RN18	**0.605**	0.587	0.580
CIFAR-100 / RN18	0.922	0.966	**0.971**
CINIC-10 / RN18	0.792	0.851	**0.858**
Location / MLP4	0.860	0.880	**0.896**
Purchase / MLP4	0.625	**0.628**	0.617
Texas / MLP4	0.805	0.834	**0.847**

Table 7: Offline AUC, $K = 8$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.677**	0.666	0.661
CIFAR-100 / WRN	**0.856**	0.847	0.851
CINIC-10 / WRN	0.691	0.749	**0.750**
Location / MLP3	0.902	0.927	**0.930**
Purchase / MLP3	**0.607**	0.596	0.588
Texas / MLP3	0.816	0.832	**0.833**
CIFAR-10 / RN18	**0.616**	0.591	0.586
CIFAR-100 / RN18	0.937	0.973	**0.974**
CINIC-10 / RN18	0.807	**0.863**	0.863
Location / MLP4	0.887	0.902	**0.906**
Purchase / MLP4	**0.634**	0.629	0.621
Texas / MLP4	0.832	0.850	**0.851**

Table 8: Offline AUC, $K = 64$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.679**	0.668	0.665
CIFAR-100 / WRN	**0.859**	0.851	0.852
CINIC-10 / WRN	0.693	**0.748**	0.745
Location / MLP3	0.906	0.928	**0.930**
Purchase / MLP3	**0.609**	0.596	0.589
Texas / MLP3	0.820	**0.832**	0.831
CIFAR-10 / RN18	**0.618**	0.592	0.588
CIFAR-100 / RN18	0.940	0.973	**0.974**
CINIC-10 / RN18	0.810	**0.864**	0.862
Location / MLP4	0.892	0.905	**0.907**
Purchase / MLP4	**0.636**	0.629	0.622
Texas / MLP4	0.835	**0.851**	0.849

Table 9: Offline AUC, $K = 254$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.073**	0.059	0.050
CIFAR-100 / WRN	**0.254**	0.247	0.247
CINIC-10 / WRN	0.149	0.152	**0.155**
Location / MLP3	0.261	**0.346**	0.320
Purchase / MLP3	0.026	**0.028**	0.025
Texas / MLP3	0.163	0.202	**0.206**
CIFAR-10 / RN18	**0.032**	0.030	0.028
CIFAR-100 / RN18	0.509	0.597	**0.599**
CINIC-10 / RN18	0.196	**0.225**	0.208
Location / MLP4	0.192	**0.226**	0.223
Purchase / MLP4	0.040	**0.043**	0.039
Texas / MLP4	0.196	0.240	**0.240**

Table 10: Offline TPR @ FPR= 0.01, $K = 8$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.087**	0.074	0.069
CIFAR-100 / WRN	0.308	**0.322**	0.320
CINIC-10 / WRN	0.123	**0.181**	0.178
Location / MLP3	0.327	**0.412**	0.405
Purchase / MLP3	0.023	**0.026**	0.023
Texas / MLP3	0.250	**0.268**	0.268
CIFAR-10 / RN18	**0.038**	0.032	0.031
CIFAR-100 / RN18	0.579	0.663	**0.664**
CINIC-10 / RN18	0.223	**0.277**	0.273
Location / MLP4	0.299	**0.333**	0.326
Purchase / MLP4	0.038	**0.042**	0.039
Texas / MLP4	0.294	**0.321**	0.316

Table 11: Offline TPR @ FPR= 0.01, $K = 64$. Offline replaces the IN-class likelihood: LiRA discards it (heuristic rescaling of the OUT-only score); BaVarIA substitutes the NIG conjugate prior, giving a principled offline attack at LiRA-equivalent per-point cost. Per-testbed mean over 32 replicates, antithetic-paired design; best per row in bold. BaVarIA-n beats LiRA on 10/12 testbeds (LiRA wins only CIFAR-10 / WRN and CIFAR-10 / RN18). Average of column means: LiRA 0.216, BaVarIA-n 0.246, BaVarIA-t 0.242.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.094**	0.082	0.080
CIFAR-100 / WRN	**0.334**	**0.334**	0.329
CINIC-10 / WRN	0.132	**0.183**	0.171
Location / MLP3	0.364	**0.421**	0.417
Purchase / MLP3	0.024	**0.027**	0.024
Texas / MLP3	0.269	**0.280**	0.278
CIFAR-10 / RN18	**0.041**	0.034	0.033
CIFAR-100 / RN18	0.611	**0.673**	0.672
CINIC-10 / RN18	0.241	**0.284**	0.279
Location / MLP4	0.327	**0.351**	0.341
Purchase / MLP4	0.041	**0.043**	0.039
Texas / MLP4	0.318	**0.333**	0.328

Table 12: Offline TPR @ FPR= 0.01, $K = 254$. Mean over 32 replicates. Best per row in bold.